What is Explainable AI, and why is it important?

Explainable AI (XAI) is the field of artificial intelligence that focuses on making machine-learning models understandable for humans. Instead of only giving a prediction, an explainable model also shows why this prediction was made and which features influenced it.

XAI is important for three main reasons:

1. **Trust:**
   Users are more willing to accept the output of a model if they can understand the logic behind it. This is especially relevant for pricing, where the result directly affects money.
2. **Transparency and fairness:**
   XAI helps detect bias, unfair patterns, or mistakes in the model. For example, if the model overvalues certain neighborhoods without a good reason, we can see it through feature explanations.
3. **Debugging and improvement:**
   Clear explanations help developers understand what the model learns and whether it uses the correct information. If the model relies on weak or irrelevant features, XAI reveals that.

There are two main ways to achieve explainability:

- **Interpretable models** (e.g., linear regression, decision trees), where the model itself is easy to understand.
- **Post-hoc explanation methods** for more complex models, such as
  Feature Importance, Partial Dependence Plots (PDP), ICE plots, SHAP values, and LIME.
  These tools help visualize how each feature affects predictions.

In pricing applications, explainability is essential because the prediction influences real financial decisions. Users need to understand why a price is considered “fair”, especially in regulated environments like Amsterdam.

What does XAI mean for this project?

Project context

The project aims to estimate a “fair” nightly price for Airbnb listings in Amsterdam using simple, static listing information (property type, room type, number of bedrooms, bathrooms, amenities, and neighborhood). The goal is to compare the predicted price to other prices and flag a listing as underpriced, normal, or overpriced.

However, during experimentation, the models achieved only R^2 around 0.2–0.3, which means that most variation in price is not explained by the available data. This is a signal that Airbnb pricing in Amsterdam depends on many dynamic factors that are not present in the dataset.

How XAI helps understand the limitations

When applying XAI tools (feature importance, PDP, SHAP), the explanations reveal several things:

- The influence of the available features is **weak and unstable**.
  Features such as the number of bedrooms, the amenities count, or the neighbourhood do affect price, but only slightly.
- PDP and SHAP plots show no strong or consistent patterns, meaning the model cannot learn meaningful price relationships.
- A large part of the prediction still comes from unexplained noise, which XAI makes visible.

These observations are important:
XAI not only explains strong models, but it also exposes when a model is unable to learn the signal.
In this project, XAI makes it clear that the missing factors (seasonality, demand, tourism peaks, events, city regulations) are responsible for most real price variation.

Role of clustering

Unsupervised clustering grouped listings into low, fair, and high-price segments relatively well. This is useful as a descriptive analytical tool, but it is not a true AI prediction model and cannot replace a supervised fair-price estimator.
It can be included as a supporting insight, but not as the final solution.

Broader domain perspective: AI pricing in short-term rentals

In the real short-term rental industry, pricing is a dynamic problem.
Actual AI pricing systems (Airbnb Smart Pricing, Price Labs, Beyond Pricing, Wheelhouse, etc.) use much more advanced data sources, such as:

- daily calendar data with booking history,
- seasonal demand patterns,
- holidays and large city events,
- competitor prices in nearby areas,
- tourism inflow and market pressure,
- city regulations and occupancy limits.

This is a very different situation compared to a static dataset of listing attributes.

Current industry trends

In the next 3–5 years, experts expect:

1. **More personalized pricing models**
   AI systems will adjust prices not only based on listing features, but also based on guest behavior and willingness to pay.
2. **Stronger regulation and transparency requirements**
   Cities like Amsterdam increasingly limit short-term rentals due to overtourism. AI pricing tools will need clear explanations to avoid unfair increases or market distortion.

Growing importance of XAI

In the short-term rental market, pricing decisions affect many people: guests, hosts, platforms, and even city authorities. Because of this, there is an increasing demand for transparent and understandable pricing algorithms**.**

XAI becomes important here because users want clear explanations of *why* a suggested price looks the way it does, especially in cities with regulations and high tourism pressure.

The goal is simple: people want to see that pricing is based on real, meaningful factors instead of random fluctuations.

Position of this project in the domain

This project is intentionally simple and static because it uses only the listing attributes available in the dataset. In reality, Airbnb pricing is influenced by many dynamic factors such as seasonality, demand, events, and policy changes. These factors are not present in the dataset, which explains the low predictive performance.

By comparing my model to the wider domain, I can clearly see:

- Listing features by themselves cannot explain most price variation.
- Real pricing systems rely heavily on dynamic and external data that my dataset does not contain.
- Therefore, my model and industry-level pricing systems operate in different contexts and serve different purposes.

Instead of being a weakness, this actually helps your argument:
My results show why simple, static models are not enough for fair-price estimation and why more advanced systems require richer data sources.

Final advice: pros, cons, and recommendations

Pros

- Clear and well-defined problem: a fair-price benchmark is useful for many stakeholders.
- The project uses interpretable models and XAI techniques, which make the system transparent.
- The work provides strong learning outcomes: exploring the limits of ML models and understanding the role of explainable AI.
- Clustering offers interesting descriptive insights into price groups.

Cons

- The dataset lacks the key drivers of Airbnb pricing (seasonality, demand, tourism flow, events, and regulations).
- As a result, all regression models achieve very low performance, meaning the predictions are not reliable for real-world use.
- There is a risk of misleading users if such predictions are presented as fair prices.
- The gap between academic data and industry practice is too large for this model to work as a real pricing tool.

Recommendation

For real-world deployment: this project should not continue in its current form (no-go). The available data is fundamentally insufficient to build a fair-price estimator.

For academic and portfolio purposes: the project should continue with a revised focus (go).
The new focus should be:

1. Showing why static listing features cannot predict real prices.
2. Using XAI to demonstrate the weaknesses and limitations of the model.
3. Positioning the project as an analysis of why real AI pricing systems need dynamic, high-signal data.
4. Presenting clustering as a secondary insight, not as a prediction model.